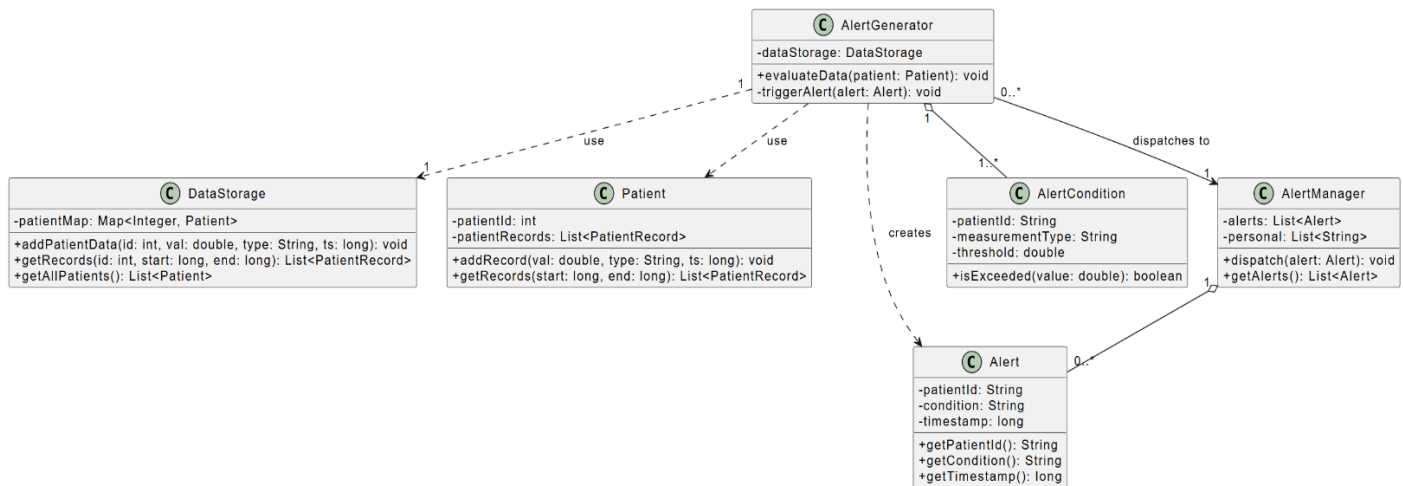




The Alert Generation System is responsible for detecting dangerous patient vitals and alerting the medical staff. The core class is **AlertGenerator** that reads patient data and compares it with some rules. These rules are stored in separate **AlertCondition** objects. This makes the system easier to update, if a threshold changes for a patient only that condition object needs to change.

If a condition is exceeded, **AlertGenerator** generates an **Alert** object that contains the patient ID, the problem detected, and a timestamp. This alert is then handed over to **AlertManager** which handles sending it to the right staff member. Separating detection and notification into separate classes means that each class has one job to do. **DataStorage** and **Patient** are connected with dashed arrows because **AlertGenerator** uses them without owning them. This keeps the alert system independent from the storage system.